D03 · Production Quality & Compliance Systems

# 02 · Quality Management Systems (ISO 9001, IATF 16949, GMP)

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

## 1 WHAT IT IS

A QMS is the documented operating system for quality — the policies, procedures, work instructions, records, and review mechanisms that together govern how the operation makes consistent, conforming product. ISO 9001, IATF 16949 (automotive), GMP (life sciences), and AS9100 (aerospace) are the dominant frameworks. The framework certifies the documentation; what determines outcomes is whether the system lives on the floor or in a folder.

## 2 WHY IT MATTERS

The QMS exists at the intersection of two worlds — the auditor's world, where documentation is the evidence of conformance, and the operator's world, where the procedure is either the way the work gets done or a paper artifact disconnected from reality. **The most expensive QMS failure mode is the system that passes audits while operations runs on tribal knowledge — because the audit pass is mistaken for system health when in fact it is theater**. The foundational discussion of the difference is in **(Juran, 1992 — Juran on Quality by Design, peer-reviewed)**: a procedure not used by the operator is a procedure that does not exist. The economic case is stark — typical plants run with documentation-to-execution gaps of **30–60% on critical procedures**, meaning the system the auditor sees is not the system the floor runs. The certification protects the contract; the gap protects the rework loop.

## 3 HOW IT IS DONE

1. **Inventory every procedure and segment by usage frequency, not by certification scope.** A typical QMS contains 800–1,500 controlled procedures, of which 100–200 are used daily by the floor, 300–500 are used quarterly by support functions, and the remainder exist only because some auditor asked for them in 2009. The classification is the first honest conversation: which procedures are operating system, which are reference, which are auditor-bait.
2. **Embed daily-use procedures into the MES, not in a document repository.** A work instruction stored as a PDF in SharePoint is a work instruction not used at the moment of work. The high-frequency procedures must live where the work happens — as electronic work instructions inside the MES, called up by the work order context. Document repositories serve auditors; the MES serves operators.
3. **Use risk-based controls, not blanket controls — IATF 16949 and ISO 9001:2015 explicitly require this.** The instinct after an audit finding is to add controls everywhere; the discipline is to add controls only where the risk justifies them. Risk assessment (FMEA-driven) determines which characteristics get SPC, which get 100% inspection, which get sampling, which get visual confirmation. Blanket controls produce procedure fatigue without commensurate quality gain **(IATF 16949:2016 + ISO 9001:2015 — international standards)**.
4. **Manage change control with rigor — the post-change defect is the most common audit finding.** Every process change (equipment, material, supplier, parameter) must trigger a documented review of the affected procedures, FMEAs, control plans, and validation status. Plants that skip this step generate the "we changed something three months ago" defects that haunt CAPA logs. PCN (Process Change Notification) discipline is non-negotiable in regulated industries and quietly essential everywhere else.
5. **Conduct management review with operational data, not with conformance summaries.** The quarterly management review required by ISO 9001:2015 §9.3 is the system's heartbeat — and the place where it most reliably degrades into ritual. The review must surface trend data on customer complaints, internal failures, audit findings, and supplier performance, and produce decisions with assigned owners and dates. A management review without decisions is paperwork.
6. **Train operators to the procedure, then audit whether they perform the procedure, then close the gap.** Training records prove training happened; gemba audits prove the procedure is followed. The closing of the gap (where observation differs from procedure) is the work — either the procedure changes to reflect reality, or the operator's practice changes to reflect the procedure, but the gap cannot persist. Living procedures or dead procedures; there is no third option.

**WORKED EXAMPLE — ILLUSTRATIVE**

A Tier 1 automotive supplier with IATF 16949 certification across 2 plants and 1,400 active controlled procedures faces escalating audit findings (47 in the 2023 surveillance audit, up from 28 in 2022) despite no major quality escapes. Root-cause analysis reveals the pattern: **312 critical procedures used daily on the floor have a documentation-to-execution gap averaging 41%** — operators do the work, but not the version in the document. The implementation: a 6-month sprint converts the 312 critical procedures to electronic work instructions inside the MES, called by work-order context; a small "shop floor documentation" team (3 FTEs) walks every procedure with operators monthly to reconcile observed practice with documented procedure. Results at audit 2024: **findings drop from 47 to 8**, operator search time for procedure drops from **3.4 minutes to 12 seconds**, change-control compliance rises from 71% to 97%, and customer PPAP rejections fall **62%**. The QMS is no longer an audit artifact — it is the operating system.

**IN PRACTICE**

### Procedure ownership is operational, not QA's

The supervisor of the work cell owns the procedure for that cell, with QE providing methodology. When operations leadership treats audit findings as operational signals rather than QA's problem, documentation stays close to the floor **(Juran, 1992 — peer-reviewed)**.

### Embed daily-use procedures in the MES, not in a document repository

Procedures live where the work happens — MES screens, work-cell terminals, mobile devices in the operator's hand. Procedures only accessible to QE and the auditor are evidence of a parallel system, not a real one.

### Audit operator practice against procedure with the goal of closing gaps

Typical plants run documentation-to-execution gaps of 30–60% on critical procedures. The audit that asks "is the document signed?" produces zero useful findings; the audit that asks "is the operator doing what it says?" produces an improvement backlog.

### Apply risk-based controls, not blanket controls

ISO 9001:2015 explicitly requires risk-based thinking. A QMS that controls everything the same way controls nothing well — and burns engineering hours on procedures that don't affect outcomes, leaving the procedures that do under-supported.

**WHAT YOU WOULD USE**

**QMS PLATFORMS**

**Veeva Vault QualityDocs / MasterControl** — The dominant platforms for regulated industries (pharma, medical device, biotech), with deep GMP/GxP compliance, validated change control, and electronic signature workflows aligned to 21 CFR Part 11 **(Veeva — vendor; MasterControl — vendor)**.

**ETQ Reliance / Sparta TrackWise / Intelex** — Broad enterprise QMS platforms that span automotive (IATF 16949), aerospace (AS9100), and general manufacturing (ISO 9001), with strong CAPA, audit, and document management modules **(ETQ — vendor; Sparta Systems — vendor)**.

**Qualio / Greenlight Guru / Arena (PTC)** — Cloud-native QMS platforms designed for small-to-mid life-sciences and medical-device operations, with faster deployment and lower total cost than the enterprise tier; the right fit when the operation is growing into regulation **(Qualio — vendor; Greenlight Guru — vendor)**.

**SharePoint + Power Automate + a disciplined librarian** — For plants without budget for a dedicated QMS platform — a structured document repository with version control, approval workflow, and a single accountable owner. Works under ISO 9001 if the discipline holds; fails the moment the librarian leaves.

## 4 SO WHAT

**The QMS that lives in SharePoint is not a system — it is a costume for the auditor.** Inventory and segment procedures by frequency of use, embed daily-use procedures in the MES, apply risk-based controls not blanket controls, manage change control with rigor, conduct management review with operational data not conformance summaries, and audit operator practice against procedure with the explicit goal of closing the gap. The framework certifies the documentation; what determines outcomes is whether the documentation is the operating system or a parallel universe accessible only to QA and the auditor.

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